

SMART SWITCH BOX

For Mobile Chargers And Modems

K. MURALI KRISHNA

Most of us leave our battery-operated devices like mobile phones, laptops, tablets, power banks and modems plugged into a charger overnight—even though these devices, especially smartphones with quick-charge option, take about two hours or less to fully charge. After the device is fully charged, the charger remains idle for more than five hours (say, from 1am to 6am). Since we usually don't access the Internet at night, modems too are just idling during this time, consuming some power.

Here we describe a power saver

circuit for mobile chargers and modems. It is a simple op-amp-based multipurpose circuit that provides individual delays for your devices. After a specific amount of time, it automatically turns the devices off. This saves power wastage during idle time while increasing the operating life-time of devices.

Block diagram of the switch box for mobile chargers and modems is shown in Fig. 1.

Circuit and working

Fig. 2 shows the switch box circuit built around op-amp IC LM339 (IC1). IC LM339 consists of four independent precision voltage comparators. These comparators are designed specifically to operate off a single power supply. Two op-amps (A1 and A2) are used in IC1.

Here the switch box is divided into two different and independent sets (Set A and Set B) to control the charging. Each set consists of a 3-pin socket, and is connected to a relay. Fuse F1 is used as a protection device.

The combination of resistor R1 and capacitor C1 in parallel forms

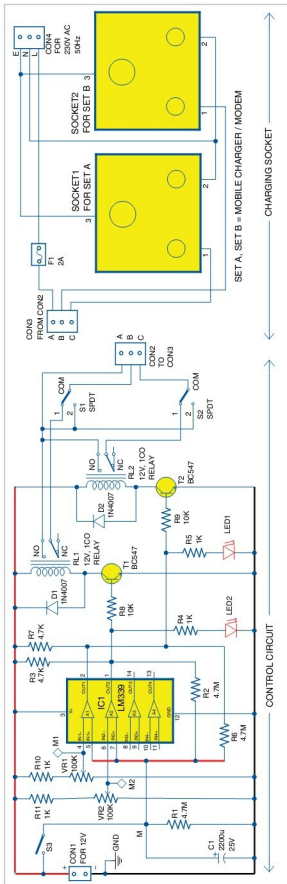


Fig. 1: Block diagram of switch box circuit

DELAY TIME FOR DIFFERENT R-C COMBINATIONS

R1 and C1	Voltage at M1/ M2	Delay in seconds
4.7 mega-ohms and 2200µF	8.47V	3600
4.7 mega-ohms and 2200µF	5.98V	7200
4.7 mega-ohms and 2200µF	4.22V	10,800
4.7 mega-ohms and 2200µF	2.98V	14,400
2 mega-ohms and 2200µF	5.20V	3600
2 mega-ohms and 2200µF	2.30V	7200
2 mega-ohms and 2200µF	1.03V	10,800
2 mega-ohms and 2200µF	0.45V	14,400

Note: Theoretical and practical values may differ because capacitor C1 discharges not only through R1 but also through op-amps and other feedback resistors.

Fig. 2: Switch box circuit